Status	Finished
Started	Friday, 16 October 2026, 6:51 PM
Completed	Friday, 16 October 2026, 6:57 PM
Duration	6 mins
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. The search is considered unsuccessful ✓
c. The first element is returned
d. An error is raised

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.
b. Linear search is more efficient than binary search on large lists.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search divides the list into halves.

Question **3**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Binary search
c. Jump search
d. Linear search ✓

Question **4**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It always starts from the beginning of the list
c. It works on unsorted lists
d. It can be applied only if the list is sorted ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Checking each element sequentially ✓
b. Sorting the list before searching
c. Skipping every second element
d. Dividing the list in half repeatedly

Question **6**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search
b. Binary search
c. Both (1) & (3)
d. Linear search ✓

Question **7**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Complexity

c. Bubble ✓

d. Selection

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element

b. By using a hash function

c. By comparing each element sequentially

d. By dividing the list length by two ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists

b. It is slower than linear search for small lists

c. It requires the list to be sorted ✓

d. It does not work with negative numbers

Question **10**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 89 and 94
c. 90 and 94
d. 90 and 99 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array
b. Item is the last element in the array or item is not there at all
c. When the item is somewhere in the middle of the array ✓
d. When the item is not the array at all

Question **12**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. At the beginning of the list
c. In the right sublist ✓
d. In the middle of the list

Question **13**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list
b. Deleting elements from a list
c. Determining whether an element is present in a list ✓
d. Inserting elements into a list

Question **14**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear
b. Both (1) & (3)
c. Binary ✓
d. Hash

Question **15**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 500
c. 1000
d. 100